

Presentation Outline

- 1 Contingency Tables
- 2 Marginal and Conditional Distributions
- 3 Mosaic Plots

Variables and Data Collection

Variables

- Variable 1
 - Number of categories = I
- Variable 2
 - Number of categories = J

Data Collection

- Sample from One Population - collect data on two categorical variables
- Multiple Samples from Multiple Populations - collect data on one categorical variable (Variable 2). Variable 1 indicates sample/population membership.

Example - Data Table

Variable 1	Variable 2
Category 1	Category 2
Category 3	Category 1
Category 2	Category 4
Category 3	Category 3
⋮	⋮
Category 3	Category 4

Contingency Table

Cross-classify observations of Variables 1 and 2.

Count number of observations in each combination and summarize in table called a Contingency Table.

- Variable 1 = Row Variable
- Variable 2 = Column Variable
- Usually include Row and Column Totals

Example - 3 X 4 Contingency Table

Variable 1	Variable 2				Total
	Cat. 1	Cat. 2	Cat. 3	Cat. 4	
Cat. 1	Y_{11}	Y_{12}	Y_{13}	Y_{14}	$Y_{1.}$
Cat. 2	Y_{21}	Y_{22}	Y_{23}	Y_{24}	$Y_{2.}$
Cat. 3	Y_{31}	Y_{32}	Y_{33}	Y_{34}	$Y_{3.}$
Total	$Y_{.1}$	$Y_{.2}$	$Y_{.3}$	$Y_{.4}$	n

Example - Eye and Hair Color

Students in an introductory statistics course were asked to select their eye and hair colors from a dropdown list.

Variable 1 = Eye Color

- Blue, Hazel/Green, Brown, Other

Variable 2 = Hair Color

- Blonde, Red, Brown, Black, Other

Example - Eye and Hair Color

Eye Color	Hair Color					Total
	Blonde	Red	Brown	Black	Other	
Blue	90	9	96	1	1	197
Hazel/Green	36	8	82	1	1	128
Brown	9	3	115	42	0	169
Other	3	1	3	3	0	10
Total	138	21	296	47	2	504

Marginal Distributions

Marginal Distributions are the distributions of Variable 1 and Variable 2 separately.

Called Marginal Distributions since values can be found in the margins of the contingency table.

Variable 1	Variable 2				Total
	Cat. 1	Cat. 2	Cat. 3	Cat. 4	
Cat. 1	Y_{11}	Y_{12}	Y_{13}	Y_{14}	$Y_{1.}$
Cat. 2	Y_{21}	Y_{22}	Y_{23}	Y_{24}	$Y_{2.}$
Cat. 3	Y_{31}	Y_{32}	Y_{33}	Y_{34}	$Y_{3.}$
Total	$Y_{.1}$	$Y_{.2}$	$Y_{.3}$	$Y_{.4}$	n

Marginal Distributions

Variable 1

- Row Totals
- Counts: $Y_{1.}, Y_{2.}, \dots, Y_{I.}$
- Proportions: $\frac{Y_{1.}}{n}, \frac{Y_{2.}}{n}, \dots, \frac{Y_{I.}}{n}$

Variable 2

- Column Totals
- Counts: $Y_{.1}, Y_{.2}, \dots, Y_{.J}$
- Proportions: $\frac{Y_{.1}}{n}, \frac{Y_{.2}}{n}, \dots, \frac{Y_{.J}}{n}$

Example - Eye and Hair Color

Eye Color	Hair Color					Total
	Blonde	Red	Brown	Black	Other	
Blue	90	9	96	1	1	197
Hazel/Green	36	8	82	1	1	128
Brown	9	3	115	42	0	169
Other	3	1	3	3	0	10
Total	138	21	296	47	2	504

Marginal Distribution of Eye Color

Eye Color	Blue	Hazel/Green	Brown	Other	Total
Counts	197	128	169	10	504
Proportions	197/504	128/504	169/504	10/504	1

Example - Eye and Hair Color

Eye Color	Hair Color					Total
	Blonde	Red	Brown	Black	Other	
Blue	90	9	96	1	1	197
Hazel/Green	36	8	82	1	1	128
Brown	9	3	115	42	0	169
Other	3	1	3	3	0	10
Total	138	21	296	47	2	504

Marginal Distribution of Hair Color

Hair Color	Blonde	Red	Brown	Black	Other	
Counts	138	21	296	47	2	504
Proportions	138/504	21/504	296/504	47/504	2/504	1

Conditional Distributions

Conditional Distributions are the distribution of one variable given a specific category for the other variable.

Distribution of Variable 2 given category of Variable 1 = Row Distribution

Distribution of Variable 1 given category of Variable 2 = Column Distribution

Variable 1	Variable 2				Total
	Cat. 1	Cat. 2	Cat. 3	Cat. 4	
Cat. 1	Y_{11}	Y_{12}	Y_{13}	Y_{14}	$Y_{1.}$
Cat. 2	Y_{21}	Y_{22}	Y_{23}	Y_{24}	$Y_{2.}$
Cat. 3	Y_{31}	Y_{32}	Y_{33}	Y_{34}	$Y_{3.}$
Total	$Y_{.1}$	$Y_{.2}$	$Y_{.3}$	$Y_{.4}$	n

Conditional Distributions

Distribution of Variable 2 given category of Variable 1.

- Values in row category i .
- Counts: $Y_{i1}, Y_{i2}, \dots, Y_{iJ}$
- Proportions: $\frac{Y_{i1}}{Y_{i.}}, \frac{Y_{i2}}{Y_{i.}}, \dots, \frac{Y_{iJ}}{Y_{i.}}$

Distribution of Variable 1 given category of Variable 2.

- Values in column category j .
- Counts: $Y_{1j}, Y_{2j}, \dots, Y_{Ij}$
- Proportions: $\frac{Y_{1j}}{Y_{.j}}, \frac{Y_{2j}}{Y_{.j}}, \dots, \frac{Y_{Ij}}{Y_{.j}}$

Example - Eye and Hair Color

Eye Color	Hair Color					Total
	Blonde	Red	Brown	Black	Other	
Blue	90	9	96	1	1	197
Hazel/Green	36	8	82	1	1	128
Brown	9	3	115	42	0	169
Other	3	1	3	3	0	10
Total	138	21	296	47	2	504

Conditional Distribution of Hair Color given Eye Color = Blue

Hair Color	Blonde	Red	Brown	Black	Other	
Counts	90	9	96	1	1	197
Proportions	90/197	9/197	96/197	1/197	1/197	1

Example - Eye and Hair Color

Eye Color	Hair Color					Total
	Blonde	Red	Brown	Black	Other	
Blue	90	9	96	1	1	197
Hazel/Green	36	8	82	1	1	128
Brown	9	3	115	42	0	169
Other	3	1	3	3	0	10
Total	138	21	296	47	2	504

Conditional Distribution of Hair Color given Eye Color = Hazel/Green

Hair Color	Blonde	Red	Brown	Black	Other	
Counts	36	8	82	1	1	128
Proportions	36/128	8/128	82/128	1/128	1/128	1

Example - Eye and Hair Color

Eye Color	Hair Color					Total
	Blonde	Red	Brown	Black	Other	
Blue	90	9	96	1	1	197
Hazel/Green	36	8	82	1	1	128
Brown	9	3	115	42	0	169
Other	3	1	3	3	0	10
Total	138	21	296	47	2	504

Conditional Distribution of Hair Color given Eye Color = Brown

Hair Color	Blonde	Red	Brown	Black	Other	
Counts	9	3	115	42	0	169
Proportions	9/169	3/169	115/169	42/169	0/169	1

Example - Eye and Hair Color

Eye Color	Hair Color					Total
	Blonde	Red	Brown	Black	Other	
Blue	90	9	96	1	1	197
Hazel/Green	36	8	82	1	1	128
Brown	9	3	115	42	0	169
Other	3	1	3	3	0	10
Total	138	21	296	47	2	504

Conditional Distribution of Eye Color given Hair Color = Blonde

Eye Color	Blue	Hazel/Green	Brown	Other	Total
Counts	90	36	9	3	138
Proportions	90/138	36/138	9/138	3/138	1

Example - Eye and Hair Color

Eye Color	Hair Color					Total
	Blonde	Red	Brown	Black	Other	
Blue	90	9	96	1	1	197
Hazel/Green	36	8	82	1	1	128
Brown	9	3	115	42	0	169
Other	3	1	3	3	0	10
Total	138	21	296	47	2	504

Conditional Distribution of Eye Color given Hair Color = Red

Eye Color	Blue	Hazel/Green	Brown	Other	Total
Counts	9	8	3	1	21
Proportions	9/21	8/21	3/21	1/21	1/21

Example - Eye and Hair Color

Eye Color	Hair Color					Total
	Blonde	Red	Brown	Black	Other	
Blue	90	9	96	1	1	197
Hazel/Green	36	8	82	1	1	128
Brown	9	3	115	42	0	169
Other	3	1	3	3	0	10
Total	138	21	296	47	2	504

Conditional Distribution of Eye Color given Hair Color = Brown

Eye Color	Blue	Hazel/Green	Brown	Other	Total
Counts	96	82	115	3	296
Proportions	96/296	82/296	115/296	3/296	1

Example - Eye and Hair Color

Eye Color	Hair Color					Total
	Blonde	Red	Brown	Black	Other	
Blue	90	9	96	1	1	197
Hazel/Green	36	8	82	1	1	128
Brown	9	3	115	42	0	169
Other	3	1	3	3	0	10
Total	138	21	296	47	2	504

Conditional Distribution of Eye Color given Hair Color = Black

Eye Color	Blue	Hazel/Green	Brown	Other	Total
Counts	1	1	42	3	47
Proportions	1/47	1/47	42/47	3/47	1

Mosaic Plot

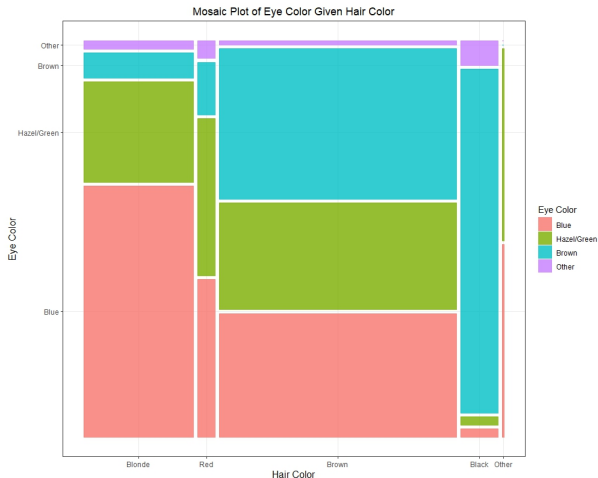
Graphical representation of one set of Conditional Distributions.

Includes representation of Marginal Distributions for conditioning variable.

Stack for each category of conditioning variable. Width of stack represents marginal distribution for conditioning variable.

Height of each color in stack represents proportion of observations in corresponding category of conditioning variable.

Example - Eye and Hair Color



Example - Eye and Hair Color

